VALVE SPRINGS

DESCRIPTION

The valve springs (Fig. 26) are made of high strength silicon chrome spring steel. The springs are common for both intake and exhaust valves.

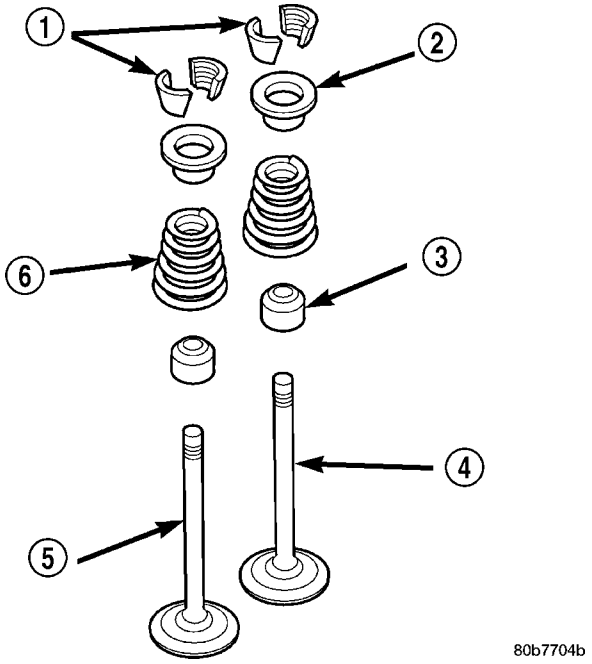

Fig. 26 VALVE AND KEEPER CONFIGURATION 4.0L

1 - VALVE LOCKS (3–BEAD)
2 - RETAINER
3 - VALVE STEM OIL SEAL
4 - INTAKE VALVE
5 - EXHAUST VALVE
6 - VALVE SPRING

STANDARD PROCEDURE - VALVE SPRING TENSION TEST

Clean all carbon deposits from the combustion chambers, valve ports, valve stems, valve stem guides and head.

Use a universal Valve Spring Tester and a torque wrench to test each valve spring for the specified tension value (Fig. 27).

Replace valve springs that are not within specifications.

REMOVAL

NOTE: This procedure can be done with the engine cylinder head installed on the block.

Inspect the valve stems, especially the grooves. An Arkansas smooth stone should be used to remove nicks and high spots.

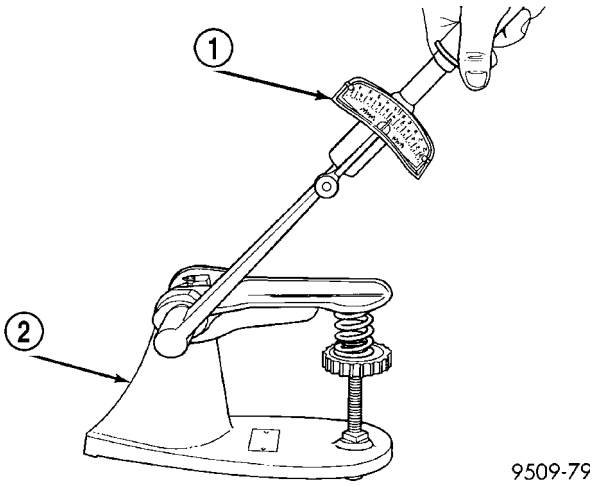

Fig. 27 Valve Spring Tester

1 - TORQUE WRENCH
2 - VALVE SPRING TESTER

Each valve spring is held in place by a retainer and a set of conical valve locks. The locks can be removed only by compressing the valve spring.

(1) Remove the engine cylinder head cover (Refer to 9 - ENGINE/CYLINDER HEAD/CYLINDER HEAD COVER(S) - REMOVAL).

(2) Remove cap screws, bridge and pivot assemblies and rocker arms (Refer to 9 - ENGINE/CYLINDER HEAD/ROCKER ARM / ADJUSTER ASSY - REMOVAL) for access to each valve spring to be removed.

(3) Remove push rods. **Retain the push rods, bridges, pivots and rocker arms in the same order and position as removed.**

(4) Inspect the springs and retainer for cracks and possible signs of weakening.

(5) Remove the spark plug(s) adjacent to the cylinder(s) below the valve springs to be removed.

(6) Connect an air hose to the adapter and apply air pressure slowly. Maintain at least 621 kPa (90 psi) of air pressure in the cylinder to hold the valves against their seats. For vehicles equipped with an air conditioner, use a flexible air adaptor when servicing the No.1 cylinder.

(7) Tap the retainer or tip with a rawhide hammer to loosen the lock from the retainer. Use Valve Spring Compressor Tool MD-998772A to compress the spring and remove the locks (Fig. 28).

(8) Remove valve spring and retainer (Fig. 28).

(9) Remove valve stem oil seals (Fig. 28). Note the valve seals are different for intake and exhaust valves. The top of each seal is marked either INT (intake/black in color) or EXH (exhaust/brown in color). DO NOT mix the seals.